Chapter 6

Quiver representations

6.1. Problems

Problem 6.1.1. Field embeddings. Recall that $k(y_1, \dots, y_m)$ denotes the field of rational functions of $y_1, \dots, y_m$ over a field k. Let $f : k[x_1, \dots, x_n] \to k(y_1, \dots, y_m)$ be an injective k-algebra homomorphism. Show that $m \geq n$. (Look at the growth of dimensions of the spaces W_N of polynomials of degree N in x_i and their images under f as $N \to \infty$.) Deduce that if $f : k(x_1, \dots, x_n) \to k(y_1, \dots, y_m)$ is a k-linear field embedding, then $m \geq n$.

Problem 6.1.2. Some algebraic geometry. Let k be an algebraically closed field, and let $G = GL_m(k)$. Let V be an algebraic representation of G. Show that if G has finitely many orbits on V, then $\dim(V) \leq m^2$. Namely:

(a) Let $x_1, \dots, x_N$ be linear coordinates on V. Let us say that a subset X of V is **Zariski dense** if any polynomial $f(x_1, \dots, x_N)$ which vanishes on X is zero (coefficientwise). Show that if G has finitely many orbits on V, then G has at least one Zariski dense orbit on V.

(b) Use (a) to construct a field embedding $k(x_1, \dots, x_N) \to k(g_{pq})$. Then use Problem 6.1.1.

(c) Generalize the result of this problem to the case when $G = GL_{m_1}(k) \times \cdots \times GL_{m_n}(k)$.